

Supplementary Materials

Passive Crowd Buffering in Pre-Operational Transit Stations: A Hybrid Simulation Study of the Thammasat University SRT Red Line Station

Author details withheld for peer review. All simulation models, climate analysis scripts, and data files supporting this study are publicly available at <https://github.com/likrentai/thammasat>

This document describes the supplementary materials supporting the above manuscript. All materials listed here will be made publicly available in a versioned open-access repository upon acceptance. Raw data, simulation model files, processed outputs, and analysis code are withheld from this submission solely to preserve author anonymity during blind review; their existence, structure, and provenance are described fully below.

1. Simulation models

1.1 Microscopic pedestrian model (AnyLogic)

A microscopic 3D pedestrian simulation was developed in AnyLogic 8 (Personal Learning Edition, v8.9.8) to cross-validate the macroscopic Machinations model against continuous-physics pedestrian dynamics. The model represents the station's Skywalk Concourse and vertical circulation spine (escalators, lifts, stairs) and was run under the off-peak baseline scenario (SC-01, $k = 0.04$) to generate the cross-validation comparison reported in Section 3.5 and Table 6 of the manuscript. The model file (.alp format) requires AnyLogic 8 or later (the free Personal Learning Edition is sufficient) to open and run. It will be deposited in full upon acceptance.

1.2 Macroscopic discrete-event model (Machinations)

A macroscopic Petri Net-based Discrete Event Simulation was developed in the Machinations platform to generate the six stress-test scenarios (SC-01 to SC-06) reported in Sections 4.1 and 4.2. The model is hosted externally on the Machinations platform. It will be publicly linked upon acceptance; the URL is withheld here to preserve author anonymity. The model represents the station as a directed graph of spatial capacity pools and throughput-constrained transitions, driven by gravity-model arrival rates. Full parameter values are reported in Tables 1, 2, and 3 of the manuscript.

2. Data sources

2.1 SRT Red Line Phase I origin-destination data

Twelve months of automated fare collection (AFC) origin-destination passenger data from the SRT Red Line Phase I network (June 2025 to May 2026) were provided by the State Railway of Thailand for academic research purposes. The dataset covers all 13×13 station pairs at hourly resolution and is anonymised and aggregated; no individually identifiable passenger information is contained in the files. Written confirmation of permission to share this dataset publicly for academic purposes has been obtained from the State Railway of Thailand. The data will be deposited in full upon acceptance. Monthly raw files are organised by Buddhist Era month code (e.g. 202506 = June 2568 BE); processed monthly OD CSVs aggregated to station-pair-hour resolution are also included as direct analysis inputs.

2.2 ERA5 reanalysis data

Monthly mean near-surface air temperature (2m temperature), dewpoint temperature, and surface pressure were obtained from the ERA5 fifth-generation atmospheric reanalysis product via the Copernicus Climate Data Store at $0.25^\circ \times 0.25^\circ$ spatial resolution, covering 1990 to 2024.

Source: Hersbach, H. et al. (2020). The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society, 146(730), 1999–2049. <https://doi.org/10.1002/qj.3803>

CDS DOI: <https://doi.org/10.24381/cds.f17050d7>

Used under the Copernicus Licence to Use Copernicus Products.

2.3 CMIP6 climate model output

Future Heat Index trajectories were derived from three CMIP6 global climate models: ACCESS-CM2, IPSL-CM6A-LR, and MPI-ESM1-2-LR, each under three experiments (historical, SSP2-4.5, SSP5-8.5) and three variables (near-surface air temperature tas, specific humidity huss, surface pressure ps). Data was obtained via the Earth System Grid Federation (<https://esgf-node.llnl.gov/projects/cmip6/>) under standard CMIP6 terms of use.

Ensemble member and grid labels: ACCESS-CM2 (r1i1p1f1, gn), IPSL-CM6A-LR (r1i1p1f1, gr), MPI-ESM1-2-LR (r1i1p1f1, gn), consistent across all experiments and variables.

Source: Eyring, V. et al. (2016). Overview of the Coupled Model Intercomparison Project Phase 6 (CMIP6). Geoscientific Model Development, 9(5), 1937–1958. <https://doi.org/10.5194/gmd-9-1937-2016>

2.4 OpenStreetMap spatial data

Land-use composition and points-of-interest within a 1 km radius of each Phase I station were extracted from OpenStreetMap via the osmnx Python library (Section 3.8 and Section 4.4.6 of the manuscript). OpenStreetMap data are made available under the Open Database Licence (ODbL).

2.5 Station architectural floor plans

Architectural floor plans used for spatial cross-checking in Section 4.2.4 are subject to redistribution restrictions and are not included in this deposit. Interested parties should request these directly from the State Railway of Thailand.

3. Python analysis workflow

Four Jupyter Notebooks implement the complete analysis pipeline. All notebooks use Python 3 with the following primary dependencies: pandas, numpy, geopandas, xarray, scikit-learn, matplotlib, osmnx, zipfile, and rarfile. Full environment specifications (requirements.txt) will be included in the public deposit upon acceptance.

Notebook	Manuscript sections	Description
OD_Data_Preparation.ipynb	Sections 3.7, 4.3	Extracts and decompresses raw OD zip files, cleans and aggregates data to station-pair-hour resolution, and extracts peak-hour k-factors for each Phase I station.
Rangsit_Profile.ipynb	Sections 3.8, 4.4	Implements k-means clustering ($k = 3$, shape-only and volume-inclusive feature sets) for station typology classification. Positions Thammasat within the resulting archetypes.
OSM_Extraction.ipynb	Sections 3.8, 4.4.6	Extracts OpenStreetMap land-use and POI data within a 1 km radius of each station using osmnx. Produces spatial catchment profiles for the proxy station comparison.
Climate_Stress_Analysis.ipynb	Sections 3.9, 4.5	Processes ERA5 and CMIP6 NetCDF files, computes relative humidity, applies the Rothfus Heat Index regression, constructs the three-model ensemble, applies a five-year rolling mean, and generates Figure 10.

4. Output files

The following processed output files are produced by the notebooks above and used directly in the manuscript analysis. They will be included in the public deposit upon acceptance.

4.1 Processed OD data

- od_202506.csv to od_202605.csv: Twelve monthly processed OD CSVs, aggregated to station-pair-hour resolution (output of OD_Data_Preparation.ipynb). These are the files used directly in the Section 4.3 empirical validation and Section 4.4 typology analysis.

4.2 Climate analysis outputs

- Per-model annual Heat Index series under SSP2-4.5 and SSP5-8.5 (six CSV files, one per model per scenario).

- Three-model ensemble mean and range under each SSP scenario (two CSV files).
- ERA5 observed annual Heat Index, raw and five-year rolling mean (two CSV files).

4.3 Manuscript figures

- Figures 1 to 10 as exported at 300 DPI (PNG format), in order of appearance in the manuscript.

5. Data availability statement

All simulation models, analysis code, processed data outputs, and raw data files described in this document will be made publicly available in a versioned open-access repository under a Creative Commons Attribution 4.0 International licence (CC BY 4.0) upon acceptance of the manuscript. The repository will be assigned a persistent DOI at that time. ERA5 data are available from the Copernicus Climate Data Store (<https://doi.org/10.24381/cds.f17050d7>) under the Copernicus Licence. CMIP6 data are available from the Earth System Grid Federation (<https://esgf-node.llnl.gov/projects/cmip6/>) under standard CMIP6 terms of use. SRT origin-destination data were provided under a research data-sharing agreement with the State Railway of Thailand; a copy of the anonymised, aggregated dataset will be included in the public deposit upon acceptance.